

Week 4 Task

Azure Storage and Azure Data Factory Pipeline

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Internship: Celebal Technologies – Data Engineering Internship

Week: Week 4

Date: 11 July 2026

Objective

The objective of this assignment is to understand Microsoft Azure cloud services and build an end-to-end data pipeline using Azure Storage Account and Azure Data Factory. The assignment includes creating Azure resources, storing data in Blob Storage, building a data pipeline, validating metadata, assigning IAM roles, and monitoring pipeline execution.

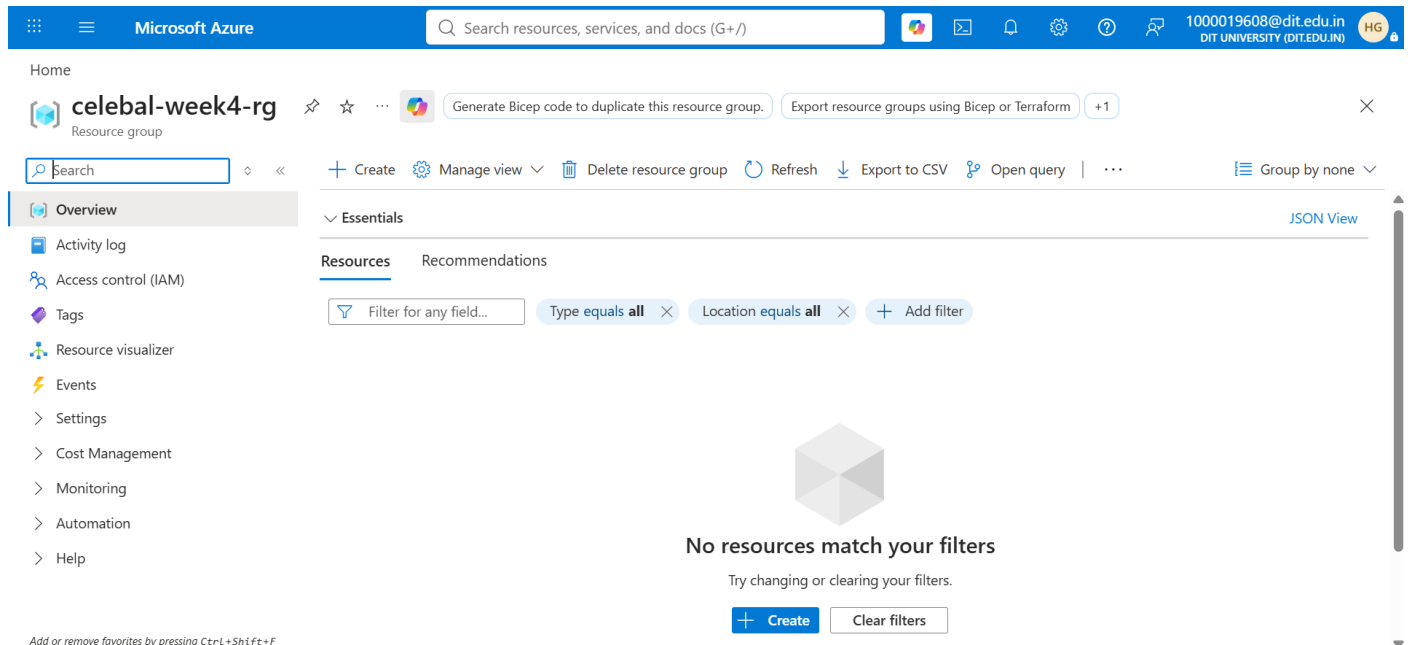
Tools and Services Used

- Microsoft Azure Portal
- Azure for Students Subscription
- Azure Resource Group
- Azure Storage Account
- Azure Blob Storage
- Azure Data Factory
- CSV Dataset (Superstore Dataset)

Task 1: Create Resource Group

A Resource Group named **celebal-week4-rg** was created in Microsoft Azure. This Resource Group will be used to organize and manage all Azure resources required for the data pipeline.

Screenshot :

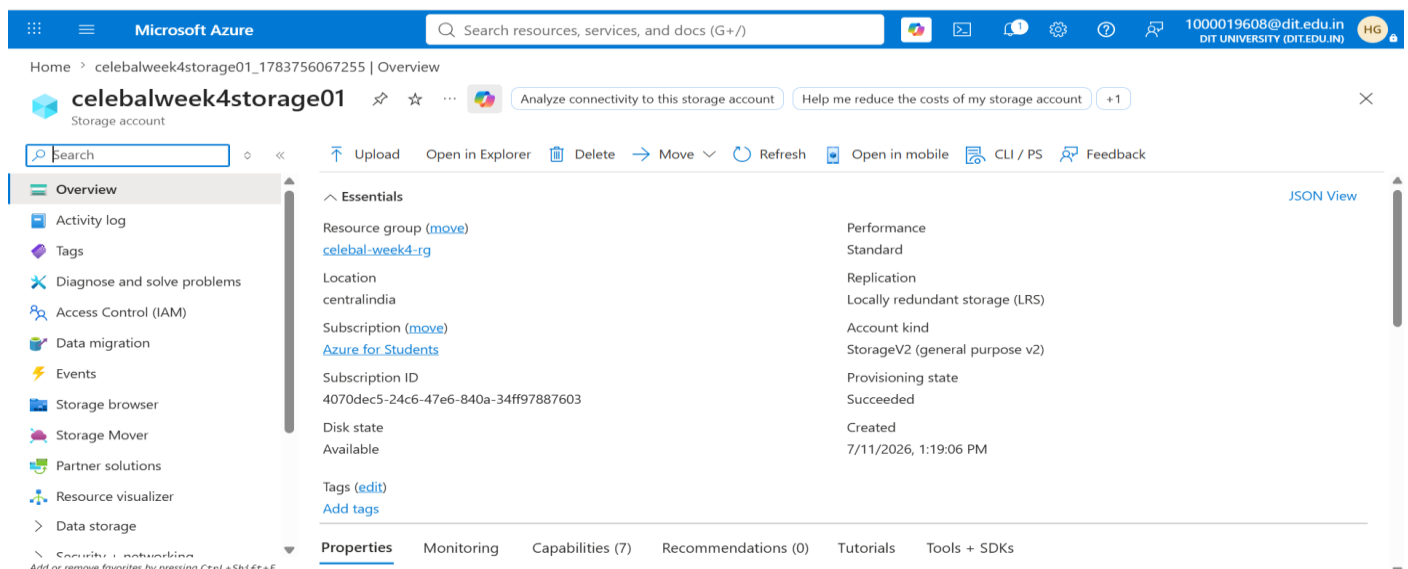


Task 2: Create Storage Account

Description:

A Storage Account was created in Azure to store the dataset required for the data pipeline. It serves as the primary storage service for uploading and managing the CSV file.

Screenshot:



Microsoft Azure

Search resources, services, and docs (G+)

Home > celebalweek4storage01_1783756067255 | Overview > celebalweek4storage01

celebalweek4storage01 | Containers

Storage account

Search

+ Add container ↑ Upload ↻ Refresh 🗑 Delete 🔒 Change access level ↶ Restore containers 🛠 Edit columns

Search containers by prefix

Only show active containers

Showing all 2 items

☐	Name	Last modified	Anonymous access level	Lease state	Storage connector	
☐	\$logs	7/11/2026, 1:19:30 PM	Private	Available	-	...
☐	input	7/11/2026, 1:29:01 PM	Private	Available	-	...

Data migration
Events
Storage browser
Storage Mover
Partner solutions
Resource visualizer
Data storage
Containers
Classic file shares
Queues
Tables
Security + networking
Data management
Add or remove favorites by pressing Ctrl+Shift+F

Microsoft Azure

Search resources, services, and docs (G+)

Home > celebalweek4storage01_1783756067255 | Overview > celebalweek4storage01 | Containers

input

Container

Search

+ Add Directory ↑ Upload 🔒 Change access level ↻ Refresh 🗑 Delete 📄 Copy 📄 Paste 🔄 Rename 🔄 Acquire lease ...

input

Authentication method: Access key ([Switch to Microsoft Entra user account](#))

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state	
☐	Sample - Superst...	7/11/2026, 1:30:28 PM	Hot (Inferred)	Block blob	2.18 MiB	Available	...

Diagnose and solve problems
Access Control (IAM)
Settings
Add or remove favorites by pressing Ctrl+Shift+F

Task 3: Azure Data Factory Basics

Objective:

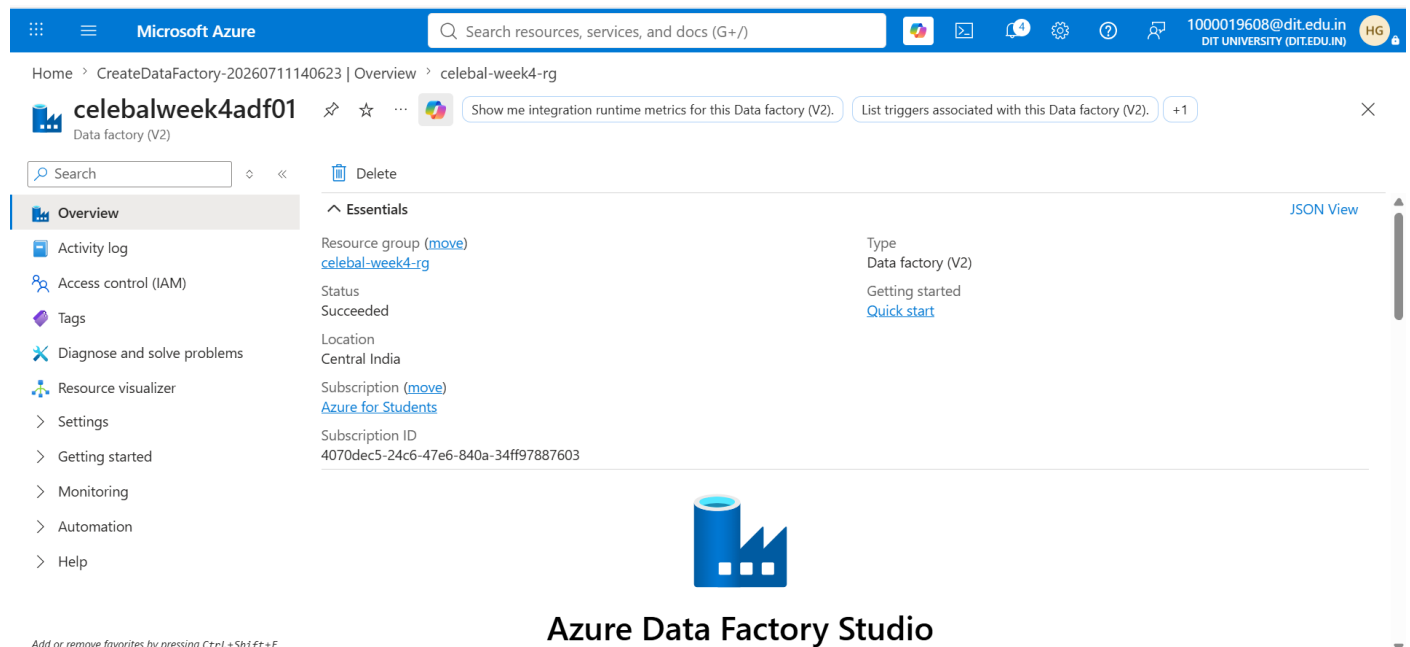
Create an Azure Data Factory instance, connect it with Azure Blob Storage, create datasets, and verify file metadata using the Get Metadata activity.

Step 1: Create Azure Data Factory

Description:

Created an Azure Data Factory instance in the Azure Portal using the existing Resource Group.

Screenshot:

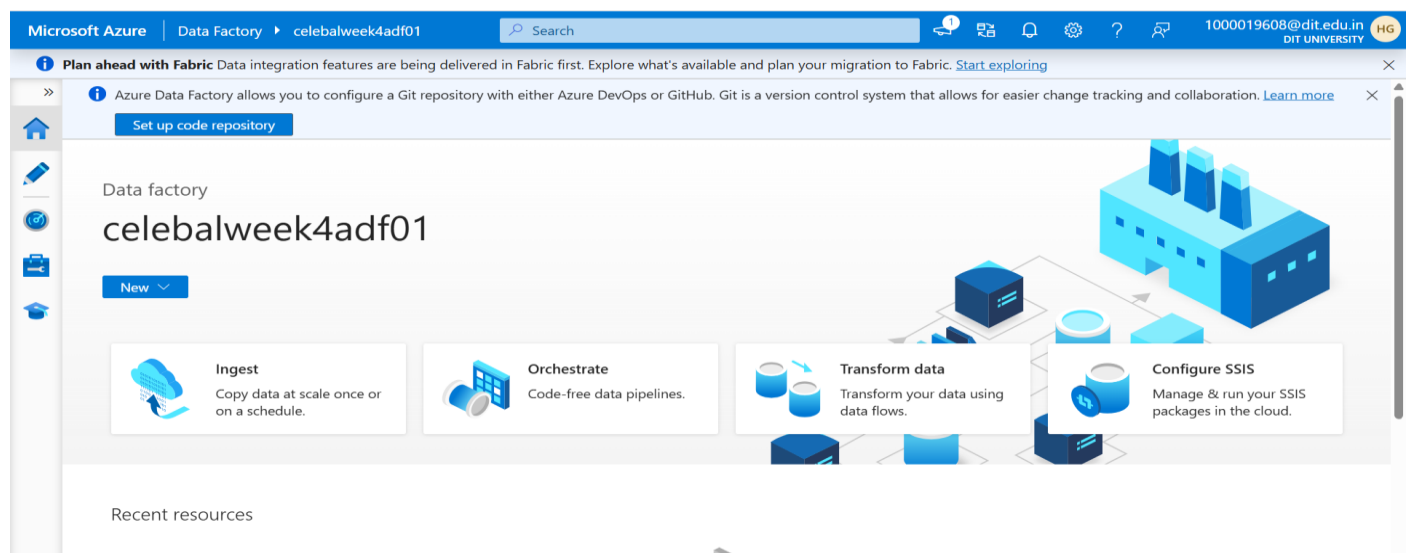


Step 2: Explore Azure Data Factory Studio

Opened **Azure Data Factory Studio** and explored:

- Author
- Monitor
- Manage

Screenshot:



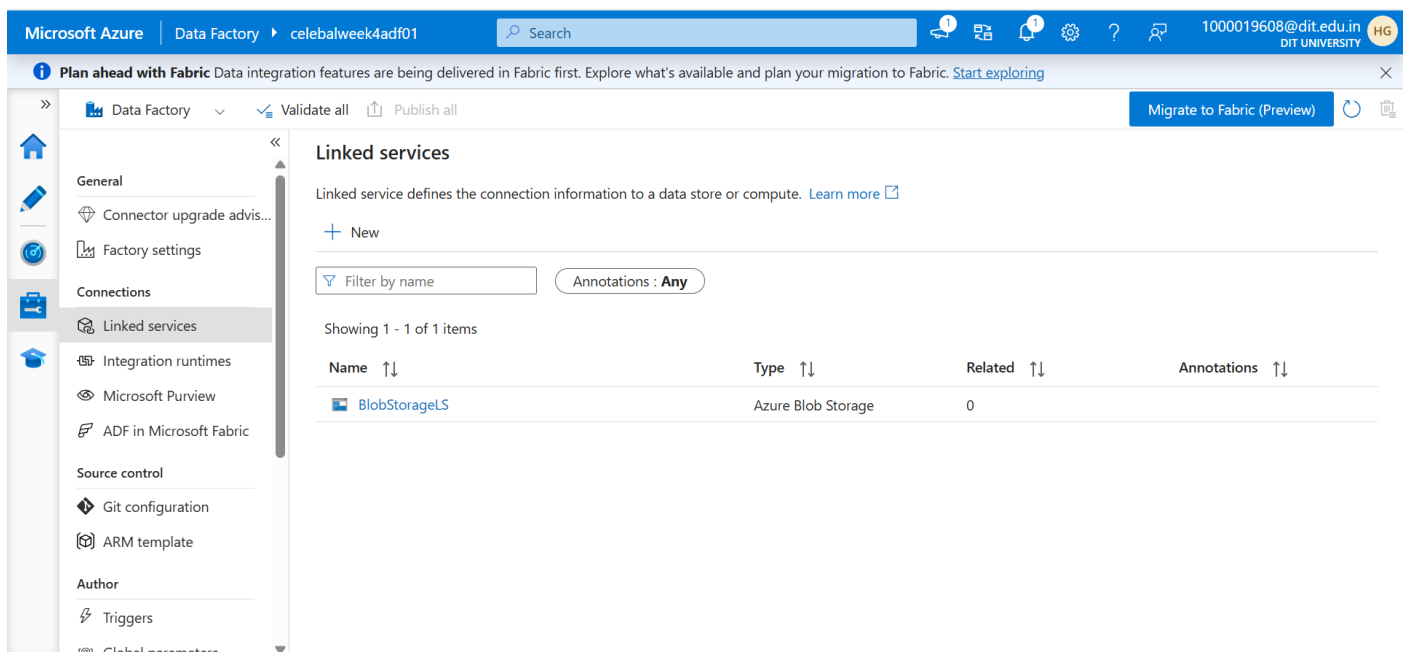
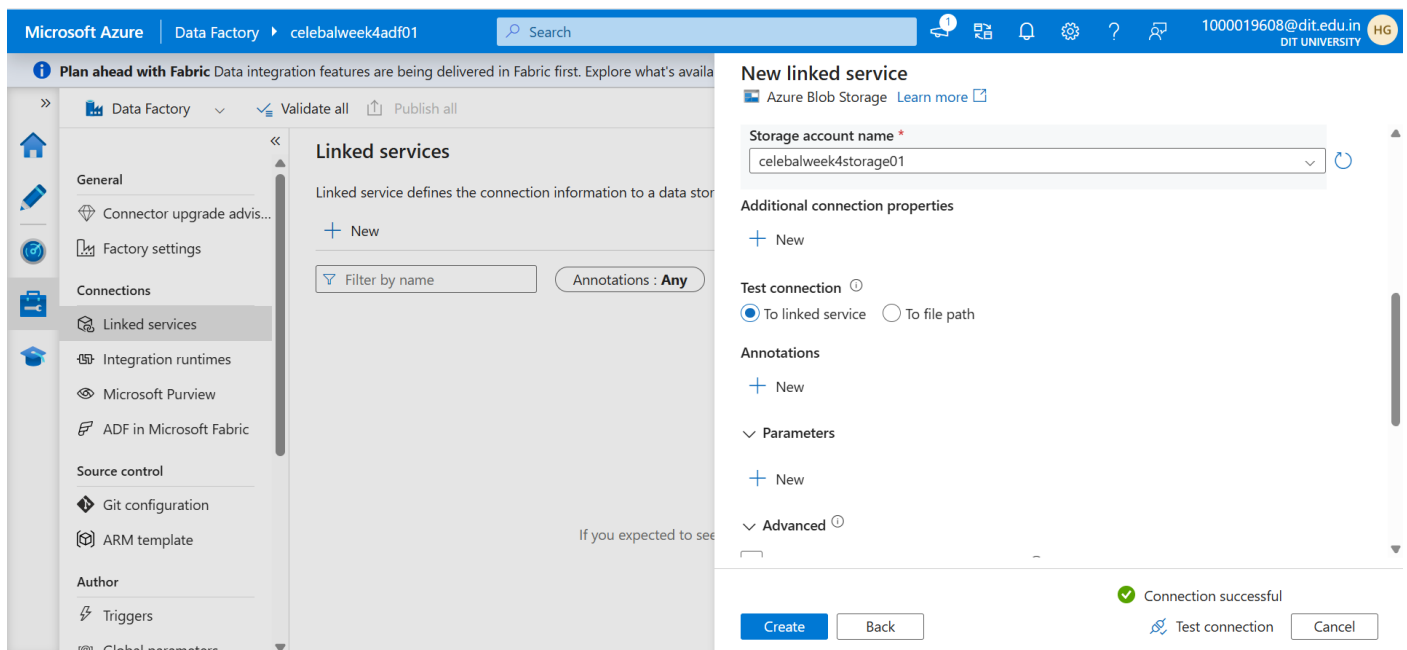
Step 3: Create Linked Service

Created a Linked Service to connect Azure Data Factory with Azure Blob Storage.

Configuration:

- Storage Type: Azure Blob Storage
- Authentication: Access Key
- Storage Account: celebalweek4storage01

Screenshot:

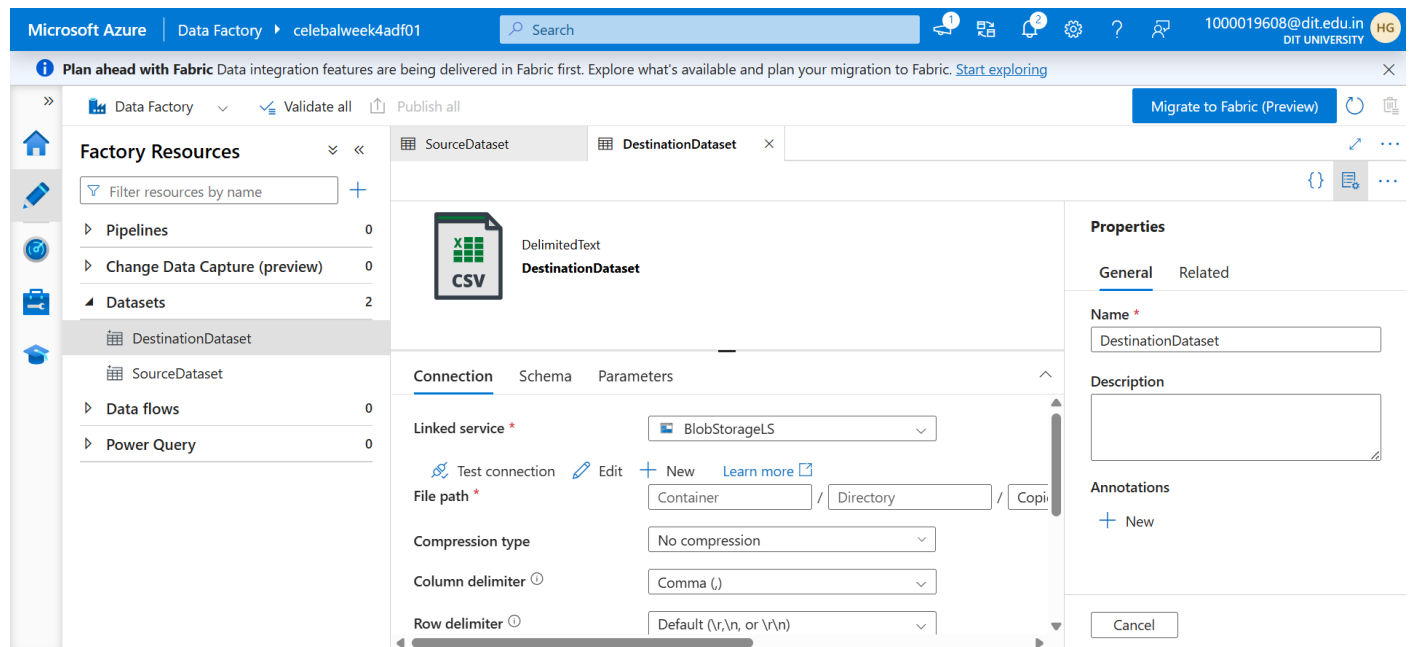
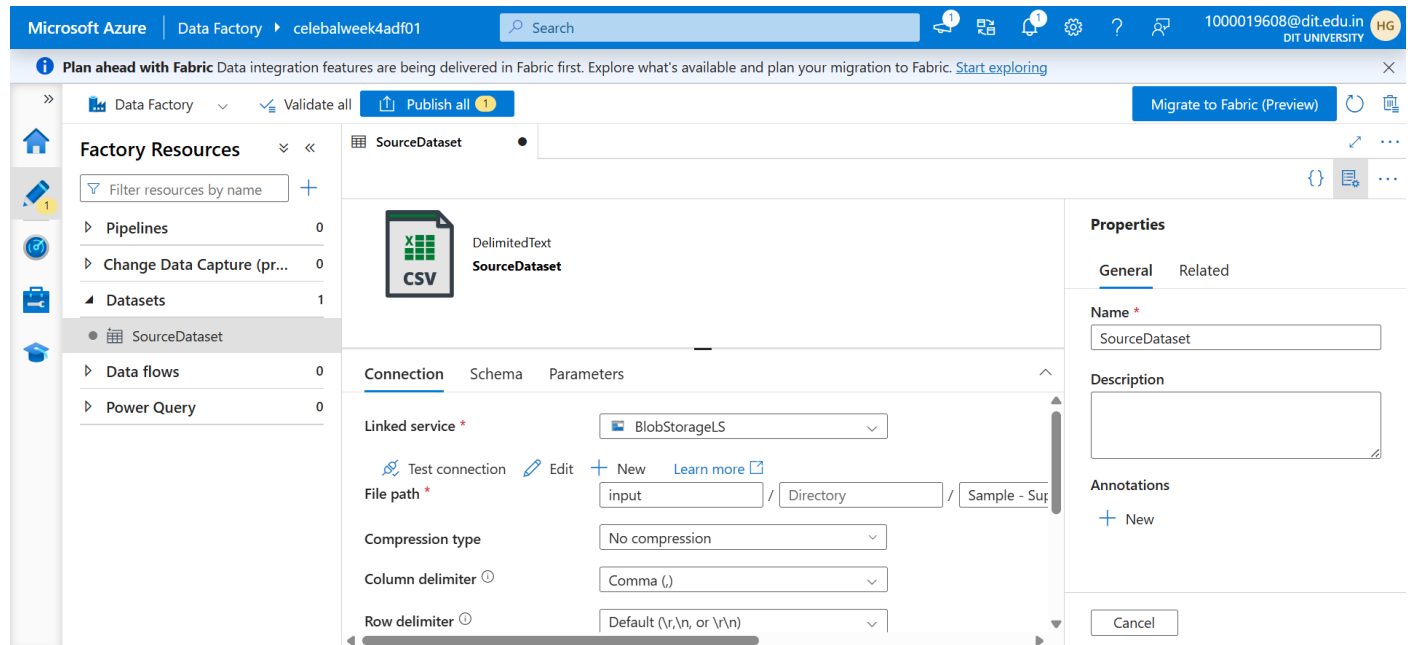


Step 4: Create Datasets

Created two datasets:

- Source Dataset (CSV file)
- Destination Dataset (Output Container)

Screenshot:



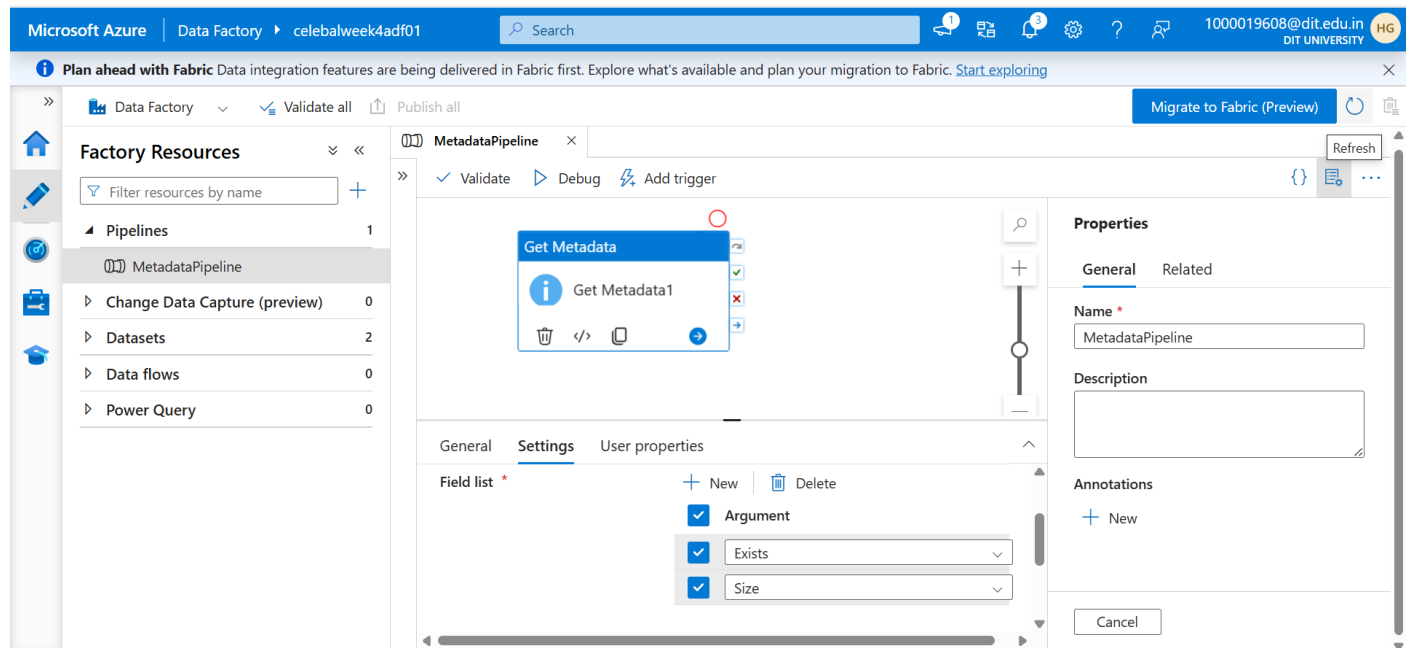
Step 5: Get Metadata Activity

Added **Get Metadata** activity inside a pipeline.

Selected Source Dataset and retrieved metadata such as:

- File Name
- Size
- Last Modified

Screenshot:



Task 4: Pipeline Development

Objective:

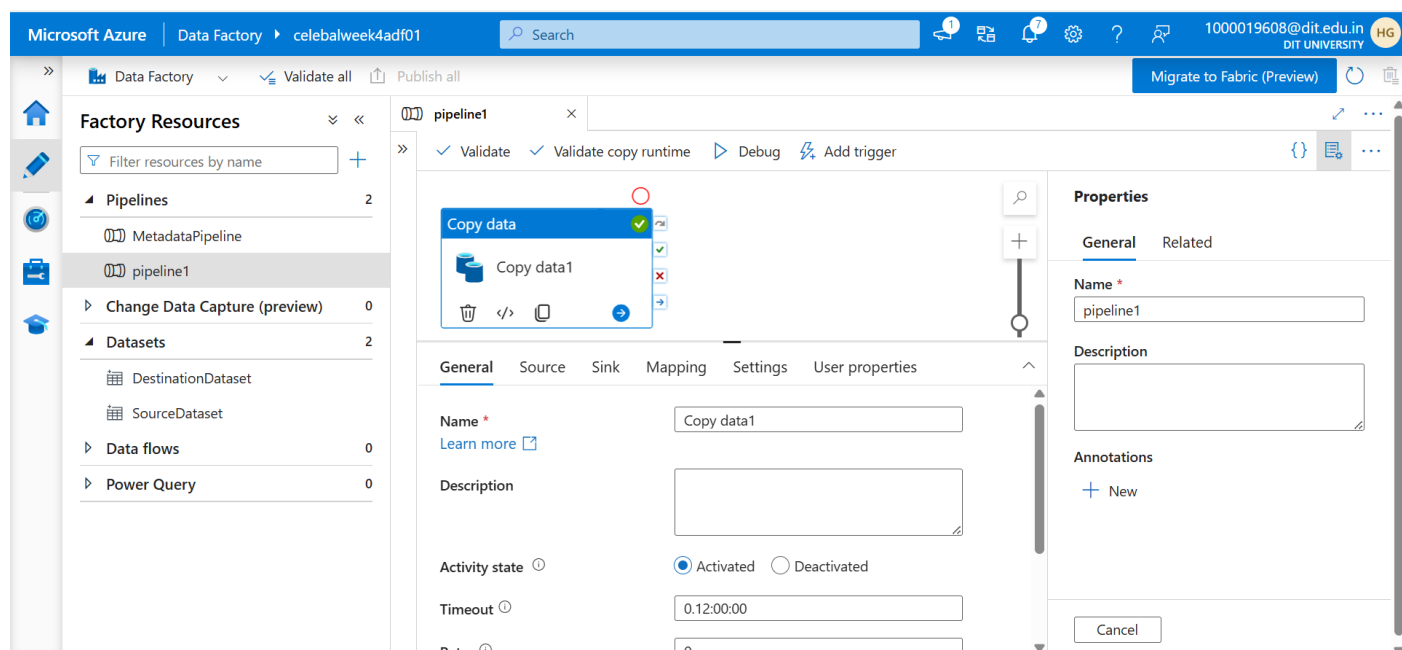
Create a data pipeline in Azure Data Factory to copy a CSV file from the source Blob Storage container to the destination Blob Storage container using the Copy Data activity.

Step 1: Create Copy Data Pipeline

Description:

Created a new pipeline in Azure Data Factory and added the **Copy Data** activity from the **Move and Transform** section.

Screenshot:



Step 2: Configure Source Dataset

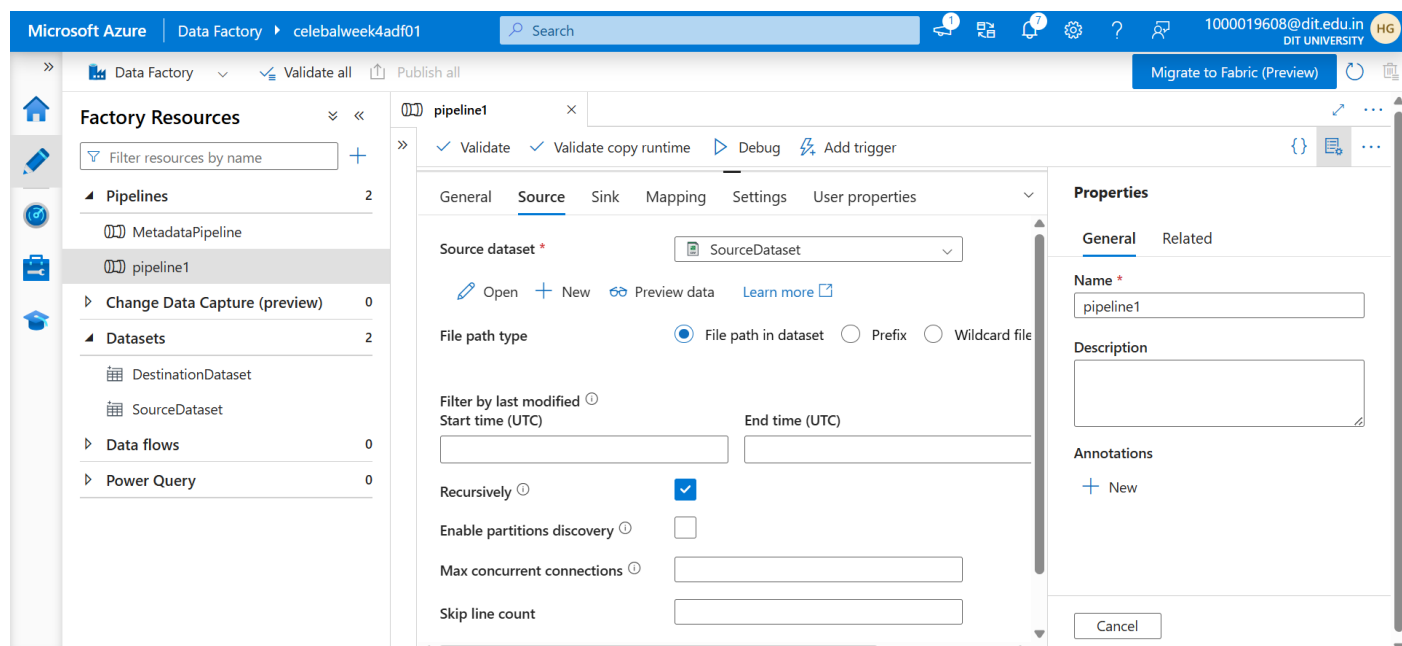
Description:

Configured the **Source** settings by selecting the previously created **Source Dataset**.

Configuration:

- Source Dataset: **SourceDataset**
- File Path Type: **File path in dataset**
- Source File: **Superstore.csv**

Screenshot:



Step 3: Configure Destination Dataset

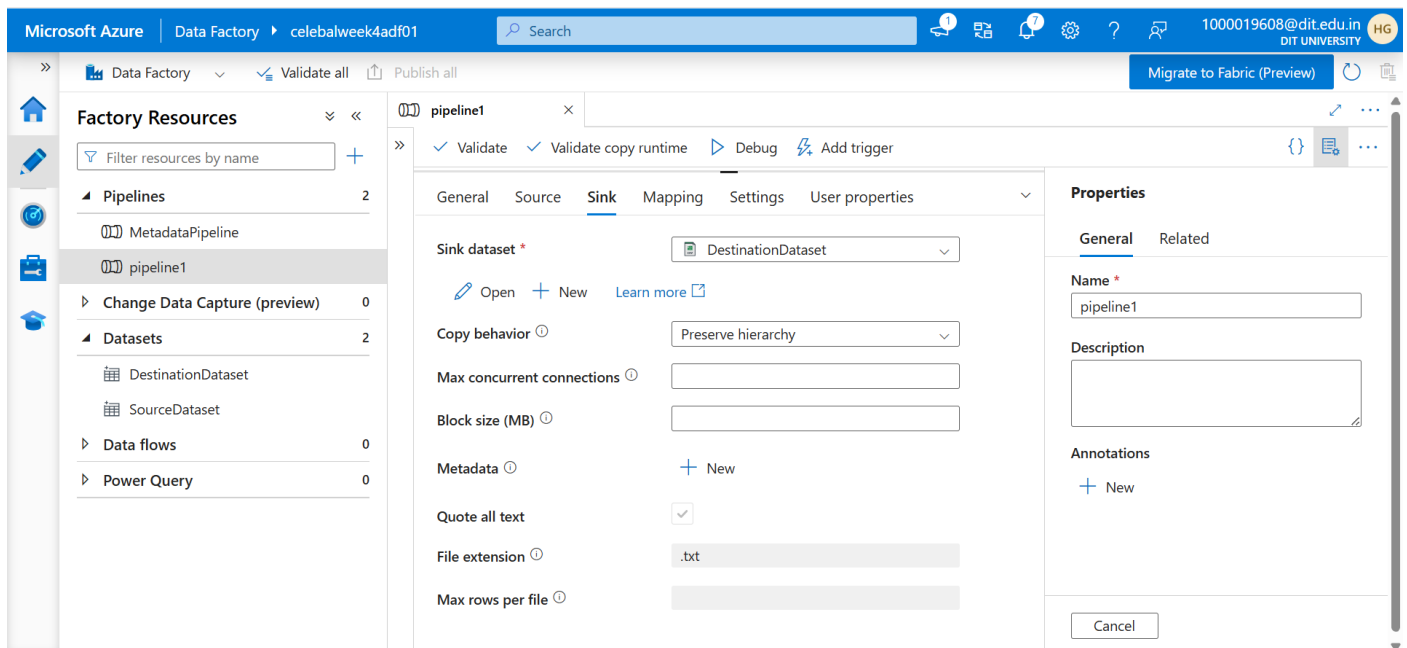
Description:

Configured the **Sink** settings by selecting the previously created **Destination Dataset**.

Configuration:

- Destination Dataset: **DestinationDataset**
- Output File: **Copied_Superstore.csv**
- Copy Behavior: **Preserve hierarchy**

Screenshot:



Task 5: Pipeline Execution

Objective:

Execute the Azure Data Factory pipeline using the **Debug** option and verify that the data is successfully copied from the source Blob Storage container to the destination Blob Storage container.

Step 1: Execute the Pipeline

Description:

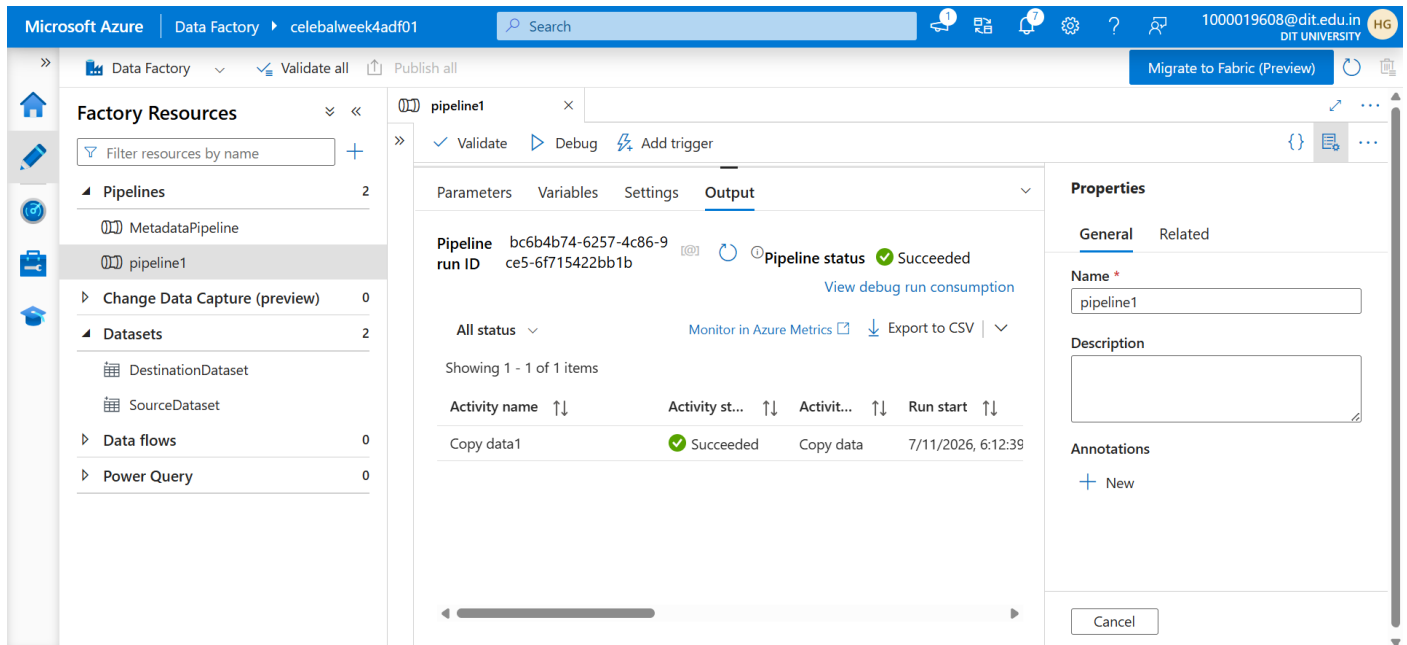
Executed the pipeline using the **Debug** option in Azure Data Factory. The pipeline copied the CSV file from the source dataset to the destination dataset.

Step 2: Verify Pipeline Execution

Description:

Verified the pipeline execution in the **Output** section. The pipeline completed successfully without any errors.

Screenshot:



Step 3: Verify Output File

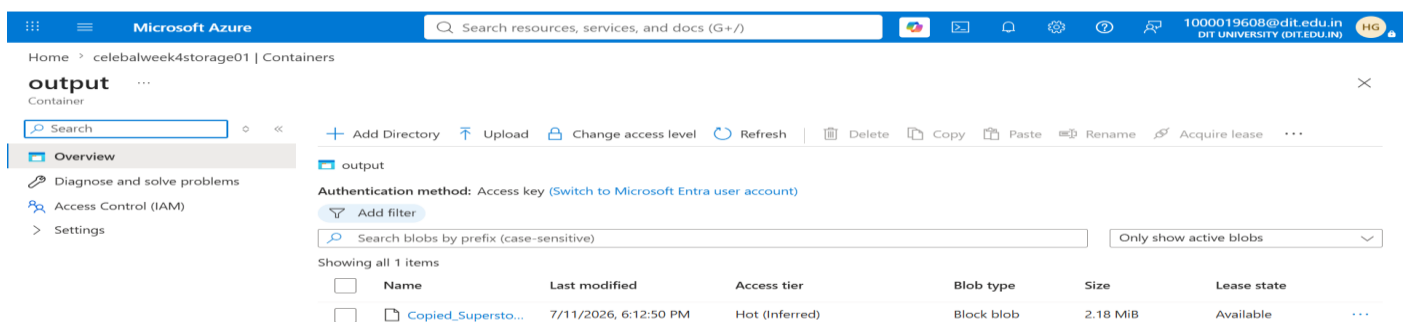
Description:

Verified that the copied CSV file was successfully created in the destination Blob Storage container.

Result:

- Source File: **Superstore.csv**
- Destination File: **Copied_Superstore.csv**
- Status: **Successfully Copied**

Screenshot:



Task 6: IAM Roles

Objective:

Assign the required Azure Role-Based Access Control (RBAC) permissions to allow Azure Data Factory to access Azure Blob Storage securely.

Step 1: Assign Reader Role

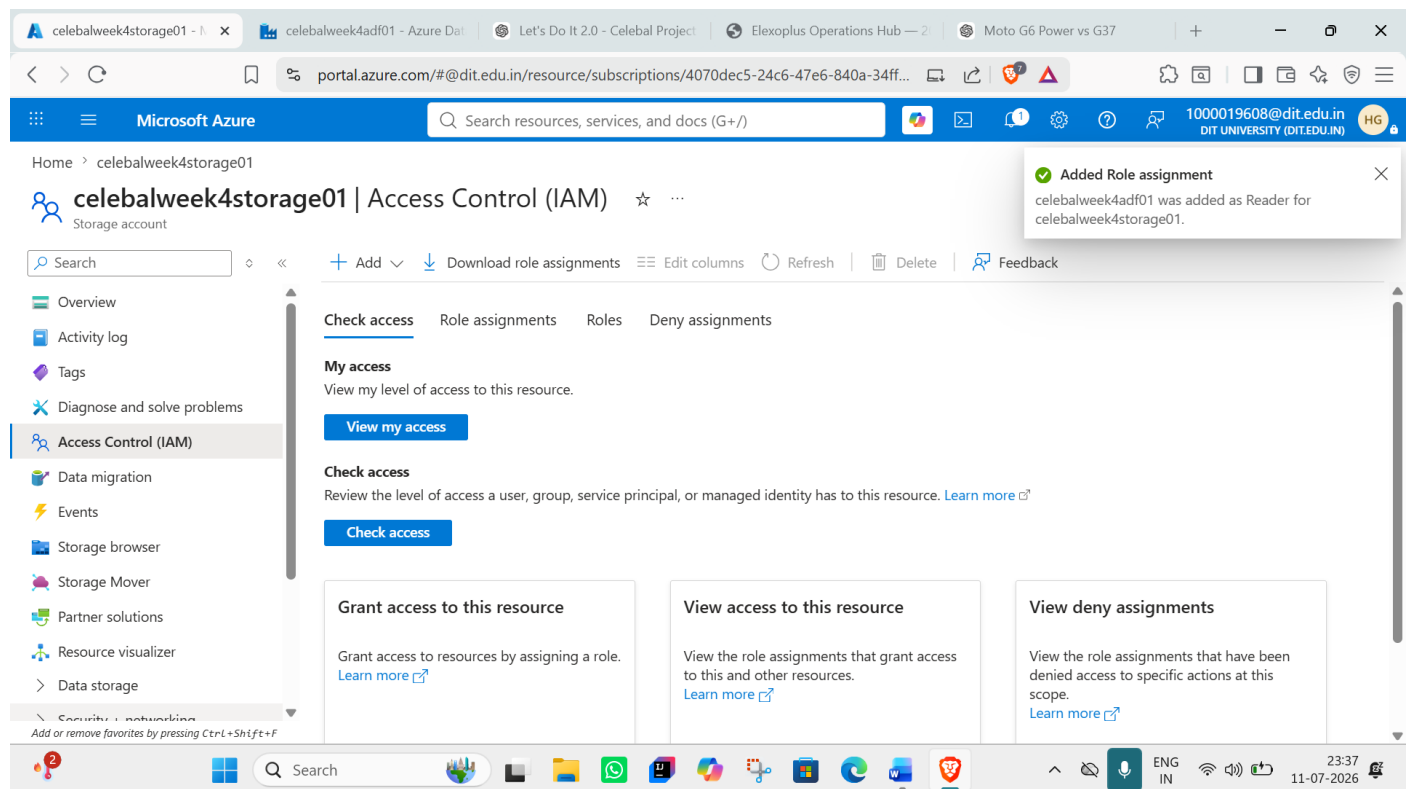
Description:

Assigned the **Reader** role to the required user/service so that it can view the Storage Account resources.

Configuration:

- Role: **Reader**
- Scope: **Storage Account**

Screenshot:



Step 2: Assign Contributor Role

Description:

Assigned the **Contributor** role to provide read and write access for managing storage resources.

Configuration:

- Role: **Contributor**
- Scope: **Storage Account**

Screenshot:

Microsoft Azure

Search resources, services, and docs (G+)

1000019608@dit.edu.in

DIT UNIVERSITY (DIT.EDU.IN)

HG

Home > celebalweek4storage01

celebalweek4storage01

Storage account

Access Control (IAM)

☆ ...

Search

◊ <<

+ Add ▾

Download role assignments

≡ Edit columns

Refresh

Delete

Feedback

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Storage Mover

Partner solutions

Resource visualizer

Data storage

Security & networking

Add or remove favorites by pressing Ctrl+L+Shift+F

Check access

Role assignments

Roles

Deny assignments

My access

View my level of access to this resource.

View my access

Check access

Review the level of access a user, group, service principal, or managed identity has to this resource. [Learn more](#)

Check access

Grant access to this resource

Grant access to resources by assigning a role. [Learn more](#)

View access to this resource

View the role assignments that grant access to this and other resources. [Learn more](#)

View deny assignments

View the role assignments that have been denied access to specific actions at this scope. [Learn more](#)

Added Role assignment

celebalweek4adf01 was added as Storage Blob Data Contributor for celebalweek4storage01.

Mini Project: Complete Data Pipeline using Azure Data Factory

Objective:

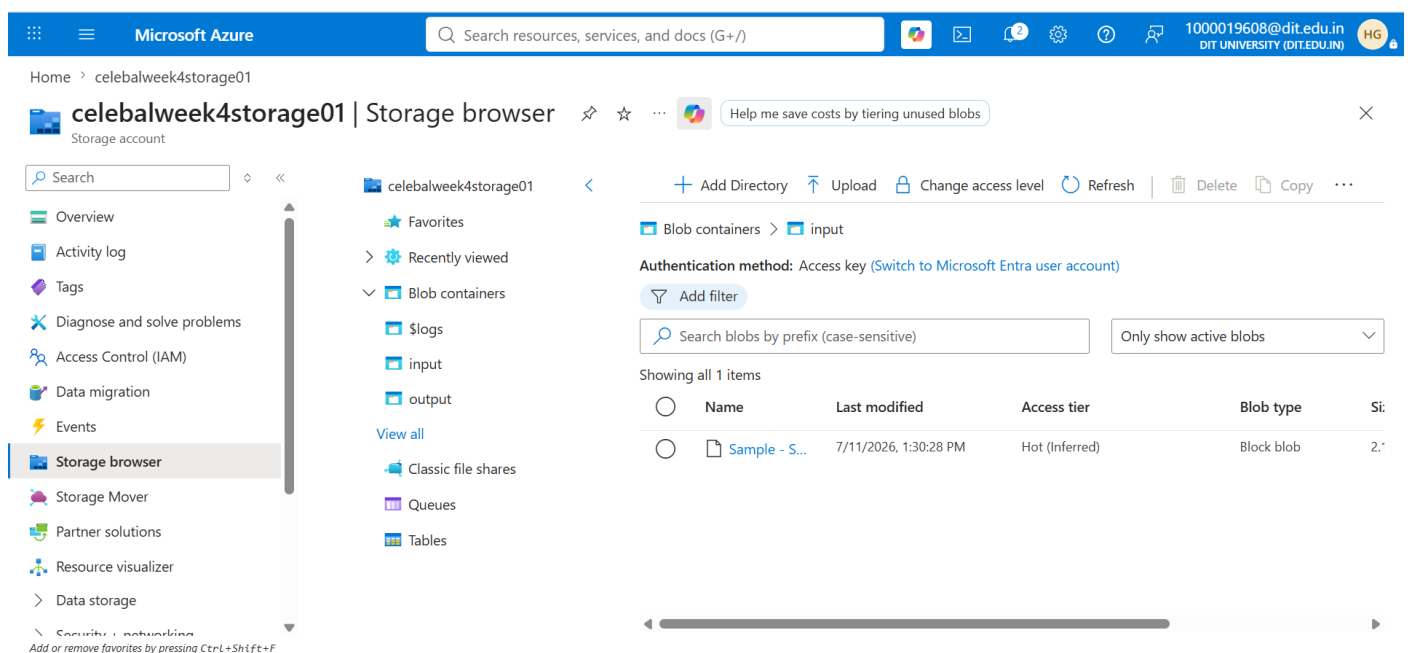
Build a complete data pipeline that reads a CSV file from Azure Blob Storage, validates file metadata, and copies the file to a destination location using Azure Data Factory.

Step 1: Verify Source File

Description:

Verified that the source CSV file was successfully uploaded and available in the Azure Blob Storage source container before executing the pipeline.

Screenshot:



Step 2: Validate File Metadata

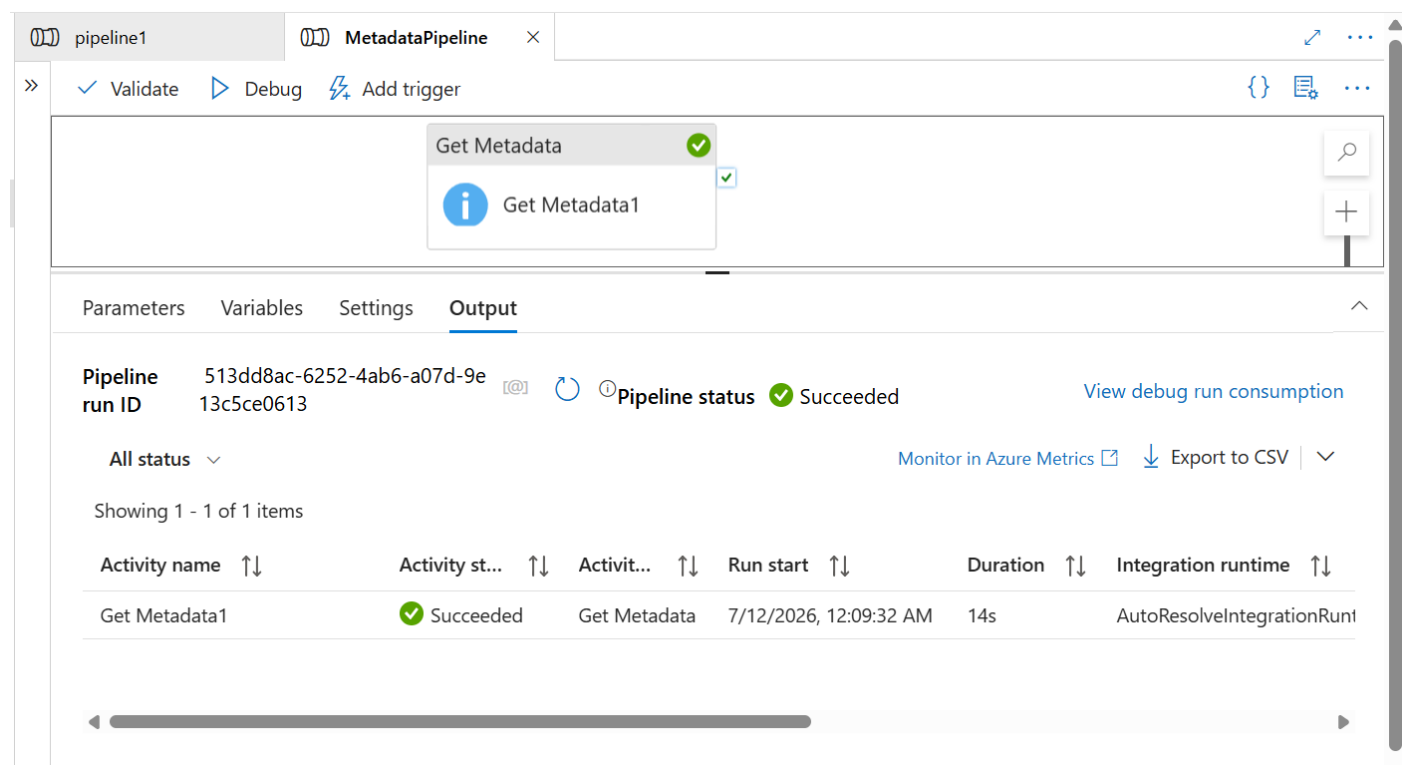
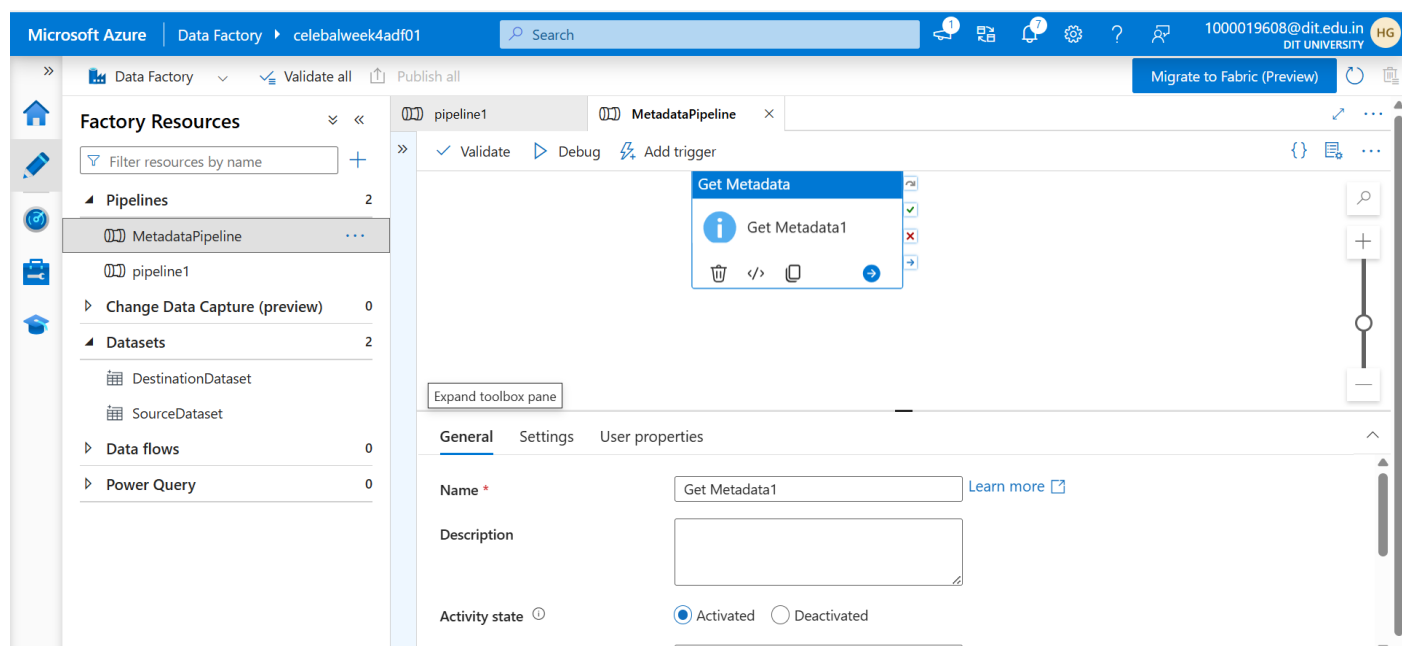
Description:

Used the **Get Metadata** activity in Azure Data Factory to validate the source file before processing.

Verified the following metadata:

- File Name
- File Size
- Last Modified

Screenshot:

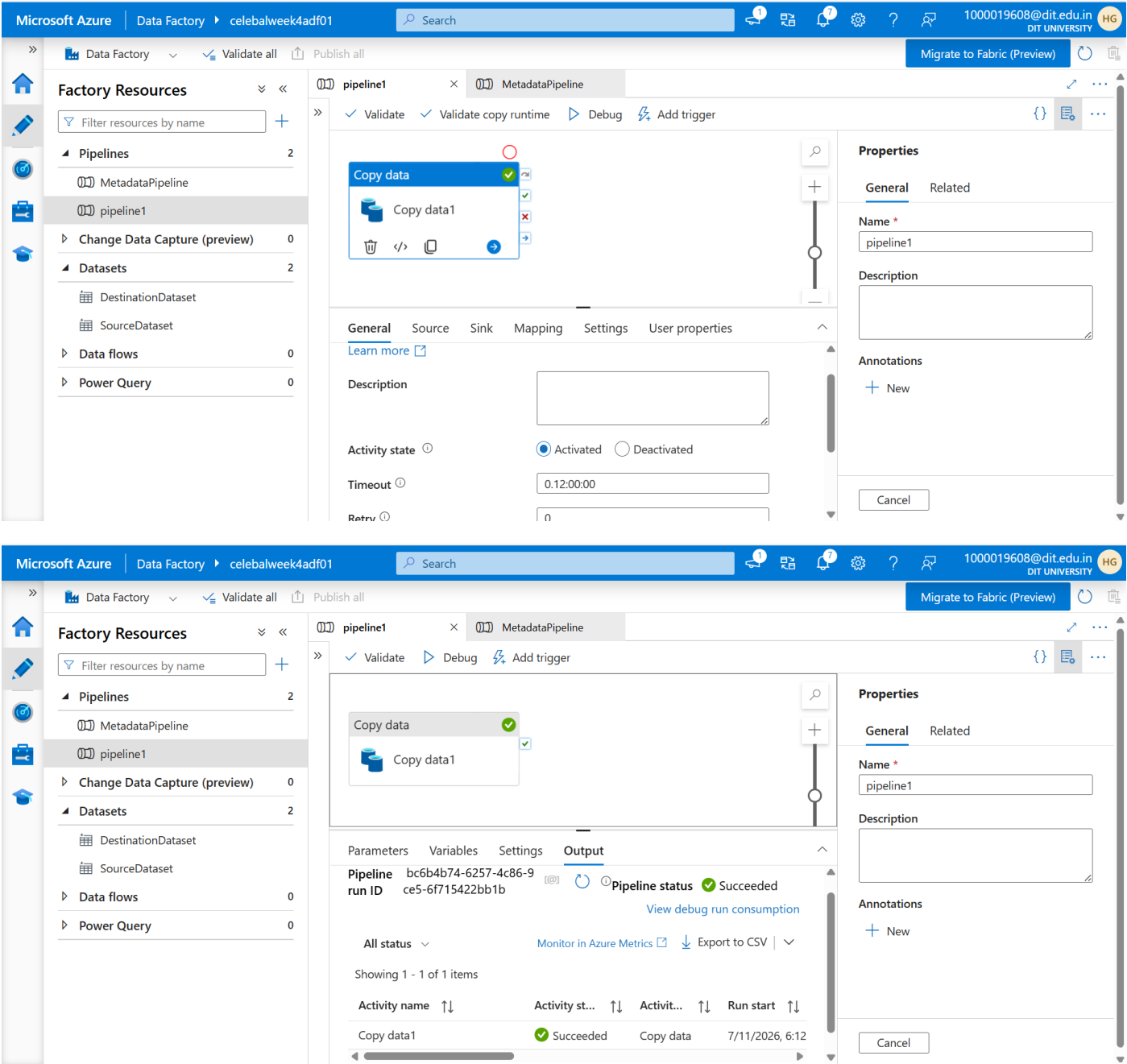


Step 3: Execute Data Pipeline

Description:

Executed the Azure Data Factory pipeline using the **Debug** option. The pipeline successfully copied the CSV file from the source Blob Storage container to the destination container.

Screenshot:



Step 4: Verify Destination File

Description:

Verified that the CSV file was successfully copied to the destination Blob Storage container after pipeline execution.

Screenshot:

The screenshot displays the Microsoft Azure Storage browser interface for the storage account 'celebalweek4storage01'. The left sidebar contains a search bar and a list of navigation options: Overview, Activity log, Tags, Diagnose and solve problems, Access Control (IAM), Data migration, Events, Storage browser (selected), Storage Mover, Partner solutions, Resource visualizer, Data storage, and Security + networking. The main pane shows the 'output' blob container. The authentication method is 'Access key'. A search bar for blobs by prefix is present. The table below shows one item:

Name	Last modified	Access tier	Blob type	Size
Copied_Sup...	7/11/2026, 6:12:50 PM	Hot (Inferred)	Block blob	2.1

Conclusion

Successfully built and executed an end-to-end Azure Data Factory data pipeline. The solution connected Azure Blob Storage using Linked Services, validated file metadata through the **Get Metadata** activity, copied the source CSV file to the destination container using the **Copy Data** activity, and verified the successful completion of the pipeline by checking the output in the destination Blob Storage.